From Static Charts to Intelligent Learning Ecosystems: A Vision for the Future of ATC/ATM Training

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Background

The U.S. air traffic controller workforce has not kept pace with growing demand, increasing workload, fatigue, delays, and operational strain.

- 13,164 controllers employed in FY2025, **6% fewer** controllers than in 2015, while **traffic grew 10%.**
- Approximately **1/3 eligible for retirement** by 2030.

Constrained ATC Training Pipeline

Lengthy training, high attrition, instructor shortages, and simulator limitations restrict how quickly new hires become certified controllers.

- **2–6 years** from application to facility certification
- Attrition occurs during screening, Academy, and facility training
- **Over 30% Academy failure rate** in FY2024

XR as Solution

Immersive technology transforms dense 2D charts into interactive 3D airspace, helps the trainees through:

- Presenting altitude and boundaries as stereoscopic 3D structures
- Allowing learners to inspect airspace from multiple viewpoints
- Connecting physical interaction with spatial concepts through embodied learning

Phase I: User Research

- Identified VFR map memorization as a major training bottleneck
- VR was preferred for experiential learning, spatial understanding, and engagement

Phase II: Prototyping & Preliminary Evaluation

- Developed a layered VFR airspace visualization tool (UE4) for self-paced exploration
- Usability evaluation with Novice population: SUS (73.2)

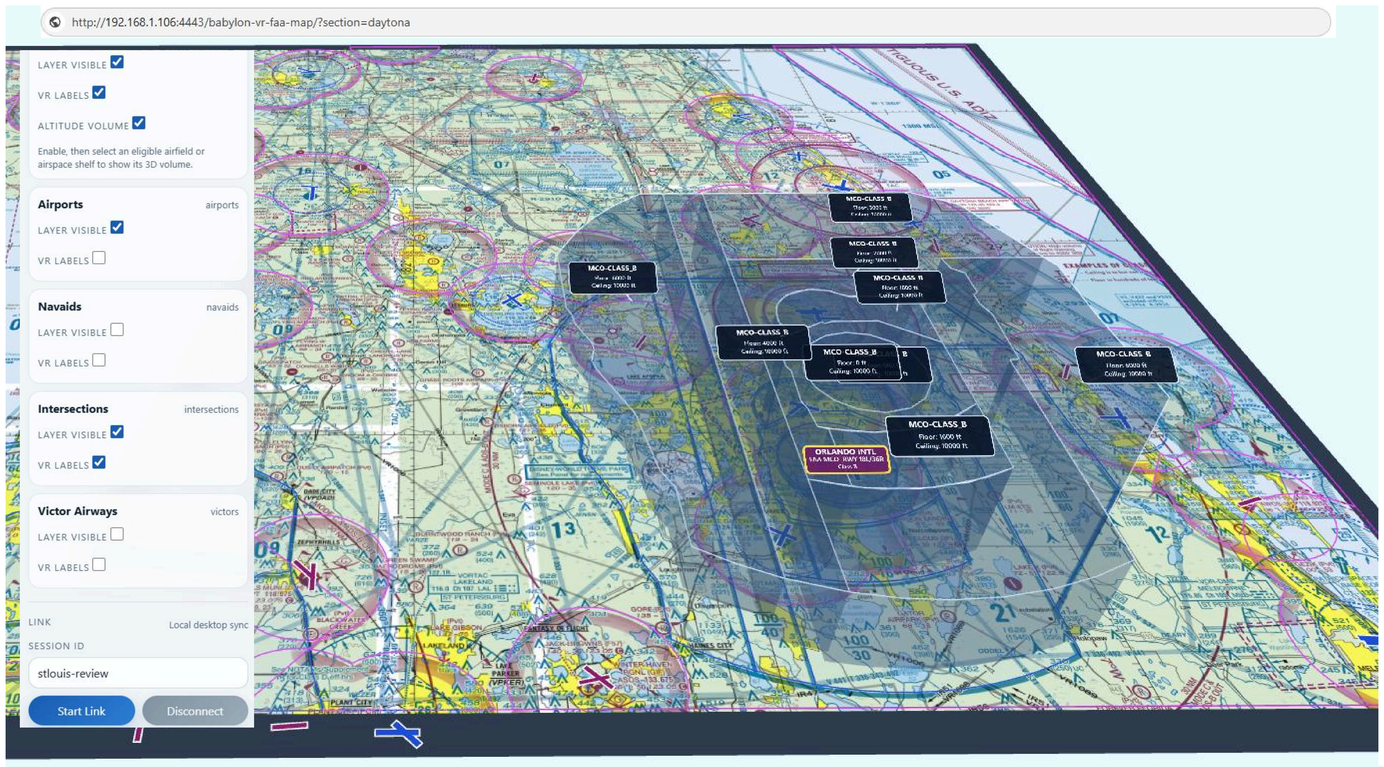
Phase III: Expert Evaluation

- Refinement based on Phase II user feedback
- Remote usability testing with CPCs: SUS(85)

Development of the User Interface and Features (Current Design)

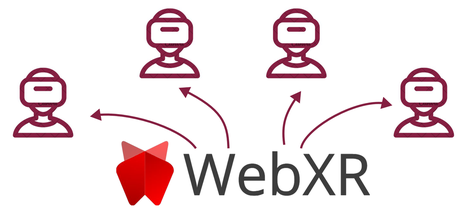
Managing Cognitive Load Through Layered Visualization

Layer toggles reduce cognitive load by helping learners isolate concepts before exploring complex airspace relationships.

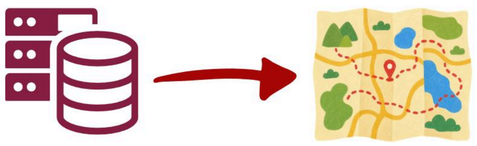


Accessible and Scalable Instruction Through WebXR

WebXR reduces access barriers while enabling synchronized, scalable instruction across desktop and immersive devices.



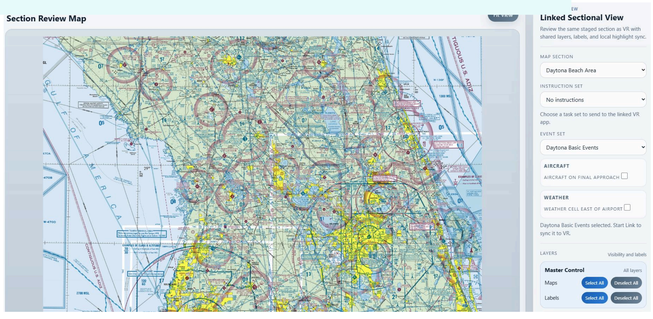
Scalable Map Generation Through a Data-Driven Pipeline



A repeatable, data-driven workflow standardizes FAA data into map layers, enabling consistent updates and scalable expansion to new regions.

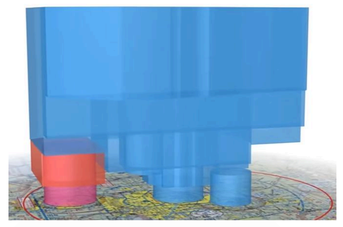
Desktop Companion

A synchronized desktop interface connects instructors and VR learners for real-time guidance, shared spatial understanding, and scalable one-to-many instruction.



3D Altitude Visualization for Spatial Learning

The altitude layer makes vertical airspace relationships visible, reducing cognitive effort and supporting spatial understanding and retention.



Discussion Questions

1. Which of these properties are necessary versus sufficient for our framework?
   - Spatial and procedural complexity
   - Risky or costly real-world practice
   - Continuous situational awareness
   - Expert-guided rehearsal
   - Difficult-to-reproduce scenarios
2. How can AI enhance safety training without replacing instructors or autonomy?
3. How spatial grounding and co-presence do immersive classroom require?
4. What should count as evidence of learning in immersive ecosystems?

Future development

AI-Personalized Training

Dynamic Event Simulation

Synchronized Instructor Tools

Next Steps

Controlled Efficacy Study

Supplemental Individual Practice

Classroom Ecosystem Integration

Funding and Partnership Opportunities

We seek funding and partners to validate and deploy our human-centered ATC training ecosystem.

- **Aviation Training Orgs**: Trainee deployment & operational testing
- **Learning & XR/HCI Researchers**: Controlled validation & research
- **Tech Partners**: Ecosystem expansion & development

FutureHCI
FUTURE OF HUMAN-COMPUTER INTERACTION WORKSHOP

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